

MertonCredit

October 1, 2026

1 The Merton Model: Reading Default Risk from Stock Prices

Question: A firm's asset value and asset volatility can't be observed, but its stock price can. If equity is a call option on the firm's assets, can we invert the option formula to recover the assets and estimate how close the firm is to default?

This notebook: 1. Prices equity as a call and debt as risk-free debt minus a put (Merton, 1974). 2. Implements three estimators: the **two-equation** solve, the **iterative KMV** procedure (Vassalou & Xing, 2004), and the **naive DD** of Bharath & Shumway (2008). 3. Checks them on **simulated firms** where the true answer is known. 4. Applies them to eight listed companies with very different leverage, both as a cross-section and as a 2023-2026 time series. 5. Maps the non-linear link between leverage, volatility and default probability, and lists where the model breaks.

Inspired by WQU MScFE 560, Module 1 (credit risk), Module 4 L3 (home equity as an option), Module 5 L3 (probability of default and credit spreads), and Module 6 L1 (balance sheet: leverage and default). References are in `reference.md`.

1.1 1. Equity is a call option on the firm

A firm has assets V and zero-coupon debt with face value F due at T . At maturity, shareholders either repay F and keep $V - F$, or, with limited liability, hand the firm to creditors. So

$$E_T = \max(V_T - F, 0), \quad D_T = \min(V_T, F) = F - \max(F - V_T, 0).$$

Equity is a **call** on the assets. Debt is **risk-free debt minus a put**. This is the same payoff as home equity with a non-recourse mortgage (Module 4). If V follows geometric Brownian motion with volatility σ_V , Black-Scholes gives

$$E = VN(d_1) - Fe^{-rT}N(d_2), \quad d_1 = \frac{\ln(V/F) + (r + \frac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}}, \quad d_2 = d_1 - \sigma_V\sqrt{T},$$

and Itô's lemma links equity and asset volatility:

$$\sigma_E E = N(d_1) \sigma_V V.$$

Distance to default is the number of standard deviations between the expected asset value and the default point. The probability of default is $PD = N(-DD)$:

$$DD = \frac{\ln(V/F) + (\mu - \frac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}}.$$

With $\mu = r$ this gives the risk-neutral PD $N(-d_2)$, which is the one that prices the debt. The model credit spread is $s = -\frac{1}{T} \ln(D/F) - r$, where $D = V - E$.

```
[1]: import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm
from scipy.optimize import fsolve

plt.rcParams.update({"figure.figsize": (10, 4.5), "axes.grid": True, "grid.
    ↪alpha": 0.3})
RNG = np.random.default_rng(1974)
DAYS = 252

[2]: def d1d2(V, F, r, sig, T=1.0):
    d1 = (np.log(V / F) + (r + 0.5 * sig ** 2) * T) / (sig * np.sqrt(T))
    return d1, d1 - sig * np.sqrt(T)

def equity_value(V, F, r, sig, T=1.0):
    d1, d2 = d1d2(V, F, r, sig, T)
    return V * norm.cdf(d1) - F * np.exp(-r * T) * norm.cdf(d2)

def distance_to_default(V, F, sig, mu, T=1.0):
    return (np.log(V / F) + (mu - 0.5 * sig ** 2) * T) / (sig * np.sqrt(T))

def credit_spread(V, E, F, r, T=1.0):
    D = V - E
    return -np.log(D / F) / T - r

# Illustration: V = 100, F = 70, r = 4%, σ_V = 25%
V0, F0, r0, s0 = 100.0, 70.0, 0.04, 0.25
E0 = equity_value(V0, F0, r0, s0)
d1, _ = d1d2(V0, F0, r0, s0)
print(f"E = {E0:.2f}    D = {V0 - E0:.2f}    _E = {norm.cdf(d1) * s0 * V0 / E0:.
    ↪1%}")
print(f"DD (risk-neutral) = {distance_to_default(V0, F0, s0, r0):.2f}    PD =_
    ↪{norm.cdf(-distance_to_default(V0, F0, s0, r0)):.2%}")
```

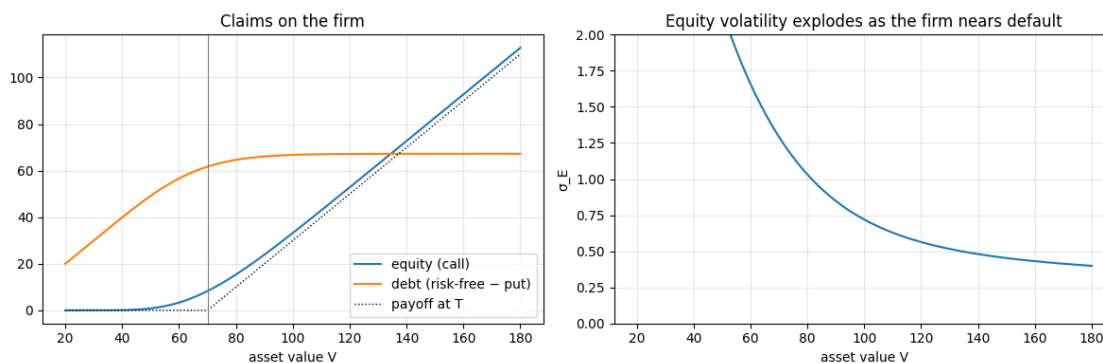
```

f"    spread = {credit_spread(V0, E0, F0, r0) * 1e4:.0f} bps")

Vs = np.linspace(20, 180, 200)
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].plot(Vs, equity_value(Vs, F0, r0, s0), label="equity (call)")
axes[0].plot(Vs, Vs - equity_value(Vs, F0, r0, s0), label="debt (risk-free - \u2192put)")
axes[0].plot(Vs, np.maximum(Vs - F0, 0), "k:", lw=1, label="payoff at T")
axes[0].axvline(F0, color="grey", lw=0.8); axes[0].set_xlabel("asset value V"); \u2192axes[0].legend(); axes[0].set_title("Claims on the firm")
axes[1].plot(Vs, norm.cdf(d1d2(Vs, F0, r0, s0)[0]) * s0 * Vs / equity_value(Vs, \u2192F0, r0, s0))
axes[1].set_xlabel("asset value V"); axes[1].set_ylabel("_E"); axes[1]. \u2192set_title("Equity volatility explodes as the firm nears default")
axes[1].set_ylim(0, 2)
plt.tight_layout(); plt.show()

```

$E = 33.23$ $D = 66.77$ $_E = 72.0\%$
 $DD \text{ (risk-neutral)} = 1.46$ $PD = 7.19\%$ $spread = 73 \text{ bps}$



The right panel is why a single “equity volatility” number is misleading. Equity volatility depends on leverage, so it changes whenever the asset value moves.

1.2 2. Three estimators

We observe the daily market value of equity E_t , the default point F and the rate r . We need V and σ_V .

Method	Idea	Weakness
Two-equation (Jones et al., 1984)	Solve the pricing and volatility equations jointly for (V, σ_V) using today’s E and a historical σ_E	Treats σ_E as constant even though leverage moves

Method	Idea	Weakness
Iterative KMV (Vassalou & Xing, 2004)	Guess $\sigma_V \rightarrow$ invert the call price for each day's $V_t \rightarrow$ re-estimate σ_V from $\ln V_t$ returns $\rightarrow$ repeat until it converges	Needs a full year of daily data
Naive (Bharath & Shumway, 2008)	$\sigma_D = 0.05 + 0.25\sigma_E$, $\sigma_V = \frac{E}{E+F}\sigma_E + \frac{F}{E+F}\sigma_D$, $V = E + F$, drift = past stock return	Not a real inversion, but forecasts default about as well

The KMV default point is **short-term debt** + $\frac{1}{2}$ **long-term debt**, with a one-year horizon.

```
[3]: def invert_assets(E, F, r, sig, T=1.0, tol=1e-10, max_iter=100):
    # Solve equity_value(V) = E for V, vectorized Newton from the right
    ↪(monotone convergence: call is convex)
    E, F, r = np.broadcast_arrays(np.asarray(E, float), np.asarray(F, float),
    ↪np.asarray(r, float))
    V = E + F
    for _ in range(max_iter):
        d1, _ = d1d2(V, F, r, sig, T)
        step = (equity_value(V, F, r, sig, T) - E) / np.maximum(norm.cdf(d1),
    ↪1e-12)
        V = np.maximum(V - step, E)
        if np.max(np.abs(step) / V) < tol:
            break
    return V

def two_equation(E, sig_E, F, r, T=1.0):
    def eqs(x):
        V, s = x
        d1, _ = d1d2(V, F, r, s, T)
        return [equity_value(V, F, r, s, T) - E, norm.cdf(d1) * s * V - sig_E *
    ↪E]
    V, s = fsolve(eqs, [E + F, sig_E * E / (E + F)])
    return V, s

def kmv_iterative(E, F, r, T=1.0, tol=1e-4, max_iter=200):
    # E, F, r: arrays of daily values over the estimation window (F may be
    ↪constant)
    E = np.asarray(E, float)
    sig_E = np.std(np.diff(np.log(E)), ddof=1) * np.sqrt(DAYS)
    F_last = np.atleast_1d(F)[-1]
    sig = sig_E * E[-1] / (E[-1] + F_last)
```

```

for it in range(max_iter):
    V = invert_assets(E, F, r, sig, T)
    lr = np.diff(np.log(V))
    sig_new = np.std(lr, ddof=1) * np.sqrt(DAYS)
    if abs(sig_new - sig) < tol:
        sig = sig_new
        break
    sig = sig_new
mu = lr.mean() * DAYS + 0.5 * sig ** 2
return {"V": V[-1], "sigma_V": sig, "mu": mu, "sigma_E": sig_E, "iters": it,
        + 1}

def naive_dd(E, F, sig_E, past_ret, T=1.0):
    sig_D = 0.05 + 0.25 * sig_E
    sig_V = E / (E + F) * sig_E + F / (E + F) * sig_D
    return (np.log((E + F) / F) + (past_ret - 0.5 * sig_V ** 2) * T) / (sig_V *
        + np.sqrt(T)), sig_V

```

1.3 3. Validation on simulated firms

Before trusting real-data estimates we test the estimators on firms where we *know* the answer. Each firm's assets follow GBM with a random $\sigma_V \in [10\%, 50\%]$ and a random leverage $F/V_0 \in [0.3, 0.9]$. The market prices equity with the true model every day, and each estimator sees only the equity series. Keep in mind that σ_V estimated from one year of daily data has sampling noise of about $\sigma_V / \sqrt{2 \cdot 252} \approx 4.5\%$ of its value, even with a perfect method.

```

[4]: def simulate_firm(sig_V, lev, mu=0.06, r=0.04, n=DAYS + 1, rng=RNG):
    V0, F = 100.0, 100.0 * lev
    z = rng.standard_normal(n - 1)
    V = V0 * np.exp(np.concatenate([[0], np.cumsum((mu - 0.5 * sig_V ** 2) /
        + DAYS + sig_V / np.sqrt(DAYS) * z))])
    return V, equity_value(V, F, r, sig_V), F, r

rows = []
for _ in range(300):
    sV, lev = RNG.uniform(0.10, 0.50), RNG.uniform(0.3, 0.9)
    V, E, F, r = simulate_firm(sV, lev)
    true_dd = distance_to_default(V[-1], F, sV, r)
    k = kmv_iterative(E, F, r)
    V2, s2 = two_equation(E[-1], k["sigma_E"], F, r)
    ndd, sN = naive_dd(E[-1], F, k["sigma_E"], r)
    rows.append({"true_sigV": sV, "lev": F / V[-1], "true_DD": true_dd,
                "KMV_sigV": k["sigma_V"], "TwoEq_sigV": s2, "Naive_sigV": sN,
                "KMV_DD": distance_to_default(k["V"], F, k["sigma_V"], r),

```

```

        "TwoEq_DD": distance_to_default(V2, F, s2, r), "Naive_DD":
        ↪ ndd})
sim = pd.DataFrame(rows)

err = pd.DataFrame({m: {"_V median abs % error": ((sim[f"{m}_sigV"] / sim.
        ↪ true_sigV - 1).abs().median()),
        "_V median % bias": (sim[f"{m}_sigV"] / sim.true_sigV -
        ↪ 1).median(),
        "DD median abs error": (sim[f"{m}_DD"] - sim.true_DD).
        ↪ abs().median(),
        "DD rank corr with truth": sim[f"{m}_DD"].corr(sim.
        ↪ true_DD, method="spearman")}}
        for m in ["KMV", "TwoEq", "Naive"])).T
err.style.format({"_V median abs % error": "{:.1%}", "_V median % bias": "{:+.
        ↪ 1%}",
        "DD median abs error": "{:.2f}", "DD rank corr with truth":
        ↪ "{:.3f}"})

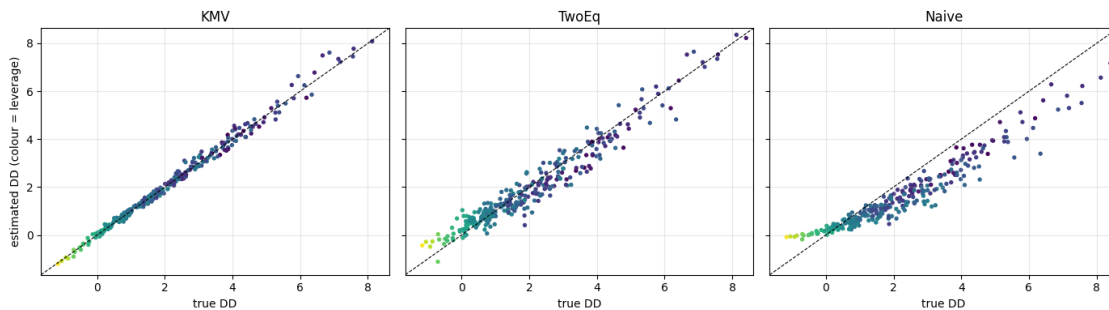
```

[4]: <pandas.io.formats.style.Styler at 0x13f6bc890>

```

[5]: fig, axes = plt.subplots(1, 3, figsize=(14, 4), sharex=True, sharey=True)
lim = [sim.true_DD.min() - 0.5, np.percentile(sim.true_DD, 98) + 0.5]
for ax, m in zip(axes, ["KMV", "TwoEq", "Naive"]):
    ax.scatter(sim.true_DD, sim[f"{m}_DD"], s=8, c=sim.lev, cmap="viridis")
    ax.plot(lim, lim, "k--", lw=0.8); ax.set_xlim(lim); ax.set_ylim(lim)
    ax.set_title(m); ax.set_xlabel("true DD")
axes[0].set_ylabel("estimated DD (colour = leverage)")
plt.tight_layout(); plt.show()

```



1.4 4. Real firms: data

Eight US firms ranging from AAA-like balance sheets to heavily indebted cyclicals:

- **Low leverage:** AAPL, MSFT, KO
- **High debt, stable:** T (telecom)

- **Cyclical / stressed:** F (autos, including its captive finance arm), INTC, AAL (airline), CCL (cruise line)

Inputs from yfinance: daily close $\times$ shares outstanding = market value of equity. The default point is **Current Debt** + $\frac{1}{2}$ **Long Term Debt** from the annual balance sheet, treated as available 90 days after the fiscal year-end (point-in-time, no look-ahead). The risk-free rate is the 13-week T-bill ($\sim$ IRX). If yfinance is unavailable, synthetic firms are generated so the notebook still runs.

```
[6]: TICKERS = ["AAPL", "MSFT", "KO", "T", "F", "INTC", "AAL", "CCL"]
REPORT_LAG = pd.Timedelta(days=90)

def load_firm(t):
    import yfinance as yf
    tk = yf.Ticker(t)
    bs = tk.balance_sheet
    if bs is None or bs.empty:
        raise ValueError(f"no balance sheet for {t}")
    get = lambda row: bs.loc[row] if row in bs.index else pd.Series(0.0,
↪index=bs.columns)
    F = (get("Current Debt").fillna(0) + 0.5 * get("Long Term Debt").fillna(0))
↪ / 1e9
    shares = get("Ordinary Shares Number") / 1e9
    fund = pd.DataFrame({"F": F, "shares": shares}).dropna()
    fund.index = pd.to_datetime(fund.index) + REPORT_LAG
    px = yf.download(t, start=str((fund.index.min() - pd.Timedelta(days=400)).
↪date()),
                    progress=False, auto_adjust=False)["Close"].squeeze().
↪dropna()
    if len(px) < 2 * DAYS:
        raise ValueError(f"too little price data for {t}")
    fund = fund.sort_index().reindex(px.index, method="ffill")
    pit = fund["F"].notna() # True once a published balance sheet exists
    # before the first report, back-fill the earliest one; such windows are
↪flagged and skipped in the history
    fund = fund.bfill()
    return pd.DataFrame({"E": px * fund["shares"], "F": fund["F"], "price": px,
↪"pit": pit})

def synthetic_universe():
    idx = pd.bdate_range("2021-01-01", "2026-09-01")
    out = {}
    for i, t in enumerate(TICKERS):
        sV, lev = 0.15 + 0.04 * i, 0.05 + 0.1 * i
        V, E, F, _ = simulate_firm(sV, lev, n=len(idx))
```

```

        out[t] = pd.DataFrame({"E": E, "F": F, "price": E, "pit": True},
                                index=idx)
        return out, pd.Series(0.04, index=idx)

try:
    import yfinance as yf
    firms = {t: load_firm(t) for t in TICKERS}
    irx = yf.download("^IRX", start="2020-01-01", progress=False)["Close"].
    squeeze().dropna()
    if len(irx) < DAYS:
        raise ValueError("no T-bill data")
    rf = np.log1p(irx / 100)
    SOURCE = "yfinance"
except Exception as exc:
    print(f"yfinance unavailable ({exc}); using synthetic firms")
    firms, rf = synthetic_universe()
    SOURCE = "synthetic"

for t, df in firms.items():
    firms[t] = df.assign(r=rf.reindex(df.index, method="ffill").bfill())
print(f"source={SOURCE}")
pd.DataFrame({t: {"start": df.index[0].date(), "end": df.index[-1].date(), "E_":
    round(df.E.iloc[-1], 1),
    "F ($bn)": round(df.F.iloc[-1], 1)} for t, df in firms.
    items()}).T

```

source=yfinance

```

[6]:
      start      end E ($bn) F ($bn)
AAPL 2021-11-24 2026-09-28 5016.6   59.5
MSFT 2022-08-24 2026-09-28 3769.2   24.8
KO   2022-02-24 2026-09-28  375.1   24.4
T    2022-02-24 2026-09-28  172.1   72.6
F    2022-02-24 2026-09-28   49.9  110.3
INTC 2022-02-24 2026-09-28  576.6   24.5
AAL  2022-02-24 2026-09-28    8.8   16.9
CCL  2022-01-24 2026-09-28   29.0   14.6

```

1.5 5. Cross-section: who is closest to default today?

Each firm is estimated with the iterative KMV method on its last 252 trading days. We report the risk-neutral DD (drift = r , which is more stable than a one-year sample drift), the physical-drift DD using the estimated μ , the naive DD, the one-year PD, and the model-implied credit spread.

```

[7]: def estimate(df):
      w = df.iloc[-DAYS:]
      k = kmv_iterative(w.E.values, w.F.values, w.r.values)

```

```

E, F, r = w.E.iloc[-1], w.F.iloc[-1], w.r.iloc[-1]
dd_rn = distance_to_default(k["V"], F, k["sigma_V"], r)
past_ret = np.log(w.price.iloc[-1] / w.price.iloc[0])
ndd, _ = naive_dd(E, F, k["sigma_E"], past_ret)
return {"E ($bn)": E, "F ($bn)": F, "leverage F/V": F / k["V"], "_E":
↪k["sigma_E"], "_V": k["sigma_V"],
        "DD (risk-neutral)": dd_rn, "DD ()": distance_to_default(k["V"],
↪F, k["sigma_V"], k["mu"]),
        "DD (naive)": ndd, "PD 1y": norm.cdf(-dd_rn),
        "spread (bps)": credit_spread(k["V"], E, F, r) * 1e4, "iters":
↪k["iters"]}

xs = pd.DataFrame({t: estimate(df) for t, df in firms.items()}).T.
↪sort_values("DD (risk-neutral)")
xs.style.format({"E ($bn)": "{:,.0f}", "F ($bn)": "{:,.1f}", "leverage F/V": "{:
↪.2f}", "_E": "{:.0%}", "_V": "{:.0%}",
                "DD (risk-neutral)": "{:.2f}", "DD ()": "{:.2f}", "DD
↪(naive)": "{:.2f}", "PD 1y": "{:.3%}",
                "spread (bps)": "{:.1f}", "iters": "{:.0f}"})

```

[7]: <pandas.io.formats.style.Styler at 0x16595c650>

1.6 6. Distance to default through time

At each month-end we re-estimate on the trailing 252 days. A window is used only if every day in it has a balance sheet that was already published (fiscal year-end + 90 days). A falling DD means the equity market is pricing in more credit risk, and it usually leads rating changes.

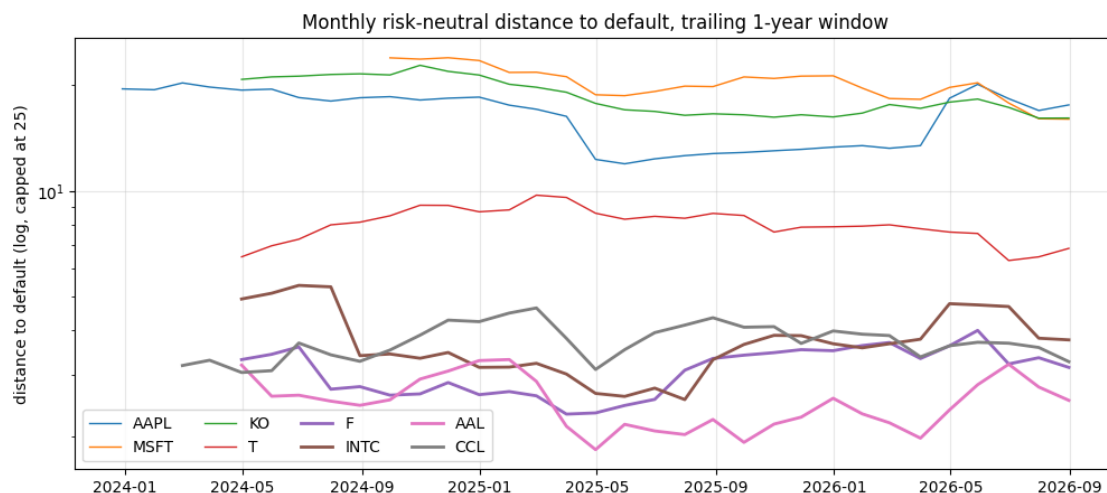
```

[8]: def dd_history(df, start="2023-01-31"):
    out = {}
    for d in pd.date_range(start, df.index[-1], freq="ME"):
        w = df.loc[:d].iloc[-DAYS:]
        if len(w) < DAYS or not w.pit.all():
            continue
        k = kmv_iterative(w.E.values, w.F.values, w.r.values)
        out[w.index[-1]] = distance_to_default(k["V"], w.F.iloc[-1],
↪k["sigma_V"], w.r.iloc[-1])
    return pd.Series(out)

hist = pd.DataFrame({t: dd_history(df) for t, df in firms.items()})
fig, ax = plt.subplots(figsize=(12, 5))
for t in hist.columns:
    ax.plot(hist.index, hist[t].clip(upper=25), label=t, lw=2 if t in ("AAL",
↪"CCL", "INTC", "F") else 1)
ax.set_yscale("log"); ax.set_ylabel("distance to default (log, capped at 25)")

```

```
ax.set_title("Monthly risk-neutral distance to default, trailing 1-year window")
ax.legend(ncol=4); plt.show()
hist.resample("YE").last().round(2).T
```



```
[8]:
```

	2023-12-31	2024-12-31	2025-12-31	2026-12-31
AAPL	19.52	18.49	13.32	17.57
MSFT	NaN	23.50	21.27	16.01
KO	NaN	21.37	16.24	16.12
T	NaN	8.72	7.90	6.86
F	NaN	2.63	3.51	3.14
INTC	NaN	3.14	3.67	3.76
AAL	NaN	3.29	2.57	2.53
CCL	NaN	4.24	3.99	3.26

1.7 7. Leverage $\times$ volatility $\rightarrow$ default probability

PD is highly non-linear in both inputs. The same asset volatility that is harmless at 30% leverage is dangerous at 80%. This is the same leverage-and-nonlinearity effect as in Module 4, seen from the lender's side. The dots place the eight firms on the map.

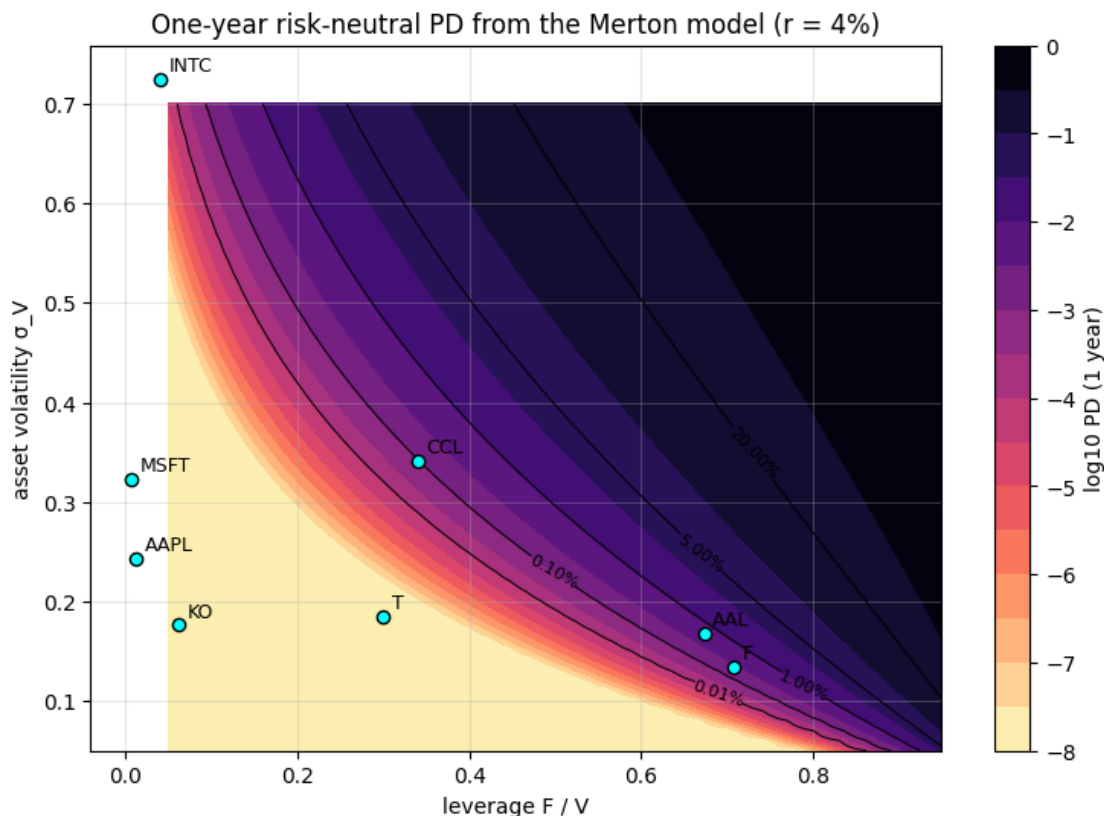
```
[9]: levs = np.linspace(0.05, 0.95, 91)
sigs = np.linspace(0.05, 0.70, 66)
L, S = np.meshgrid(levs, sigs)
PD = norm.cdf(-distance_to_default(1.0, L, S, 0.04))

fig, ax = plt.subplots(figsize=(9, 6))
cs = ax.contourf(L, S, np.log10(np.clip(PD, 1e-8, 1)), levels=np.arange(-8, 0.
    ↪5, 0.5), cmap="magma_r")
lines = ax.contour(L, S, PD, levels=[1e-4, 1e-3, 1e-2, 0.05, 0.2], colors="k",
    ↪linewidths=0.7)
```

```

ax.clabel(lines, fmt=r"%1.2f", fontsize=8)
ax.scatter(xs["leverage F/V"], xs["_V"], c="cyan", edgecolor="k", zorder=3)
for t, row in xs.iterrows():
    ax.annotate(t, (row["leverage F/V"], row["_V"]), xytext=(4, 4),
        textcoords="offset points", fontsize=9)
fig.colorbar(cs, label="log10 PD (1 year)")
ax.set_xlabel("leverage F / V"); ax.set_ylabel("asset volatility _V")
ax.set_title("One-year risk-neutral PD from the Merton model (r = 4%)")
plt.show()

```



1.8 8. Takeaways

The numbers below come from the run with yfinance data up to Sep 2026.

- **The option view works as a measuring device.** On simulated firms, the iterative KMV estimator recovers asset volatility to within about 4% (close to the sampling-noise floor for one year of data) and ranks distance to default almost perfectly (Spearman 0.997). The two-equation shortcut is about $3\times$ less accurate because it freezes equity volatility while leverage moves. The naive `_V` overstates asset volatility by about a third, yet it still ranks firms reasonably (0.97). That is why Bharath & Shumway found it forecasts almost as well.
- **The ranking makes sense.** AAL (leverage 0.67), Ford (0.71) and CCL are closest to default

with DD = 2.5–3, and AAPL, MSFT and KO are 15+ standard deviations away. INTC shows why volatility matters as much as leverage: debt is tiny relative to equity, but a 79% equity volatility pulls its DD down to about 4.

- **The levels are too optimistic: the credit spread puzzle.** The model gives AAL a 1-year PD of about 0.6% and a spread of about 3 bps. Its bonds actually trade at hundreds of bps over Treasuries. Pure-diffusion Merton cannot produce meaningful short-horizon spreads because default can't happen by a sudden jump, and it ignores liquidity and risk premia (Huang & Huang, 2012). Practitioners therefore map DD to empirical default frequencies (KMV's EDF) instead of reading $N(-DD)$ literally.
- **The balance-sheet input is the weakest link.** Ford's default point includes about \$100bn of Ford Credit debt that is backed by loan receivables, not by the auto business. So "leverage 0.71" overstates the risk of the industrial firm. Banks and captive-finance firms need a different default point.
- **Relative changes are what matter.** The monthly DD history is most useful as an early-warning signal. A falling DD for one firm, relative to its own history and its peers, contains information that annual ratings update slowly.

Extensions: map DD to realized default rates, add jumps (Zhou, 2001) or a stochastic default barrier (Black-Cox) to fix short-term spreads, compare model spreads with market CDS/bond spreads, and use DD as a factor in the equity cross-section (Vassalou & Xing, 2004).